

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.5: Moments of Equilibrium — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.5: Moments of Equilibrium
Driving Question	DRIVING QUESTION: Why do some objects stay balanced while others topple — and how can we use the physics of turning forces to predict and control whether an object will be stable or fall?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Why Did the Stool Fall? — Introducing the Moment of a Force	A three-legged Jiko stool with a sufuria of ugali at its centre remained stable, but toppled when a full 20-litre jerrycan was placed at one corner and nudged. A wider, lower four-legged stool with the same jerrycan simply rocked and self-righted. In the beam-balance activity, moving a mass further from the pivot caused greater turning, and two different masses could balance when placed at specific distances on opposite sides.	A force acting at a distance from a pivot creates a MOMENT (turning effect). The moment of a force is calculated as $M = F \times d$ (force multiplied by perpendicular distance from the pivot) and is measured in Newton-metres (N·m). The jerrycan created a large moment about the stool's pivot leg because of its large weight AND its distance from that leg — not weight alone.	The stool toppled because the jerrycan's weight (200 N) acting at a large perpendicular distance from the pivot leg created a large clockwise moment ($M = F \times d$) that was larger than the opposing (anticlockwise) moment of the stool's own weight acting at its centre of gravity. The four-legged stool did not topple under the same conditions — suggesting that something about the geometry (width of base, position of pivot legs) affects whether the turning effect causes permanent toppling or is resisted. This 'something' — involving centre of gravity and conditions for equilibrium — remains to be explained in Lessons 2 and 3.
Lesson 2 Turning Forces: Introducing the Moment of a Force	Students observed that moving a 500 g mass further from the pencil-pivot on a metre rule increased the tipping effect, even though the mass (and therefore the force) did not change. They also observed, in a teacher demonstration, that pushing a door near the hinge requires far more effort than pushing near the outer edge. In the anchoring phenomenon, they re-observed	Students learned that the turning effect of a force is called its MOMENT, and it is calculated using the formula $M = F \times d$, where F is the applied force in Newtons and d is the perpendicular distance from the force's line of action to the pivot, in metres. The SI unit of moment is the Newton-metre (N·m). A larger force OR a larger perpendicular distance produces a larger	Students can now partially explain the anchoring phenomenon: the Jiko stool topples when the jerrycan is at the corner because the perpendicular distance from the jerrycan's weight (force) to the nearest supporting leg (pivot) is large, producing a large moment (turning effect). When the jerrycan is near the centre, the distance d is small, so the moment is small and the stool

	that placing the jerrycan at the corner of the Jiko stool (far from the supporting legs) produced a dramatic toppling effect that placing it near the centre did not.	moment. This explains why the jerrycan at the corner of the Jiko stool produces a larger toppling moment than the same jerrycan placed at the centre.	remains upright. What students cannot yet explain: what happens when there are turning forces acting in OPPOSITE directions (clockwise and anticlockwise) — this is the Principle of Moments, which is the focus of Lesson 3.
Lesson 3 Lesson 3: Centre of Gravity and the Tipping Point — Why Does the Jiko Stool Fall?	Students observed that a plumb-line hanging from different points on an irregular cardboard shape always intersects at the same point (the centre of gravity), and that a tall narrow block tipped when a mass was moved to its edge while a wide low block did not — directly mirroring the jerrycan-on-stool behaviour from the anchoring phenomenon.	Students learned that every object has a centre of gravity — the single point through which its total weight appears to act — and that a body topples when the vertical line through its centre of gravity passes outside its base of support; they also distinguished between stable, unstable, and neutral equilibrium using this principle.	Students explained why the Jiko stool topples when the jerrycan is placed at the corner: the jerrycan raises and shifts the combined centre of gravity of the stool-plus-load system until the vertical line through the CG falls outside the triangular base of support formed by the three legs, generating an unbalanced toppling moment with no restoring moment to counteract it.
Lesson 4 The Principle of Moments: Balancing Forces on a Bao Plank	Learners balanced a metre-rule on a pencil pivot using bags of beans and coins, recording mass and distance data across 6+ trials and noting whether the rule was balanced or tipping.	When a ruler (or any beam) is balanced, the clockwise moment (force $\times$ distance) always equals the anti-clockwise moment about the pivot — this is the Principle of Moments. The position of a load relative to the pivot determines whether a turning effect causes toppling or stability.	The Jiko stool topples when the jerrycan is placed far from the centre because its clockwise moment about the tipping edge exceeds the anti-clockwise moment of the stool's own weight acting through the centre of gravity. The wider stool has a larger anti-clockwise stabilising moment, so the same jerrycan load does not cause toppling — the moments remain balanced.
Lesson 5 How Do Engineers Keep Things From Falling? — Factors Affecting Stability and Real-Life Applications	When clay was added to the TOP of the tall box, the critical tipping angle was small (the box toppled easily). When clay was moved to the BASE, the critical tipping angle was much larger (harder to topple). When cardboard flaps widened the base, the critical tipping angle increased further. In the case study, the crane toppled when its load was swung beyond the rated radius, shifting the effective centre of gravity outside the base.	Stability increases when the centre of gravity is lowered and when the base of support is widened. An object topples when the vertical line through its centre of gravity falls outside the base of support. Engineers apply both strategies deliberately in crane design, double-decker bus geometry, silo construction, and load distribution.	In the Jiko stool phenomenon, the narrow three-legged stool toppled when the jerrycan was placed at the corner because the combined centre of gravity of stool-plus-load shifted far to one side, and the tall load kept the centre of gravity high — so only a small nudge moved the vertical line outside the narrow base. The wide four-legged stool survived because its broad base meant the vertical line had to travel much further before exiting the base of support, even with the jerrycan at the corner.
Lesson 6 Can We Explain and Predict Toppling? — Consolidation, Model Finalisation, and DQB Review	In the demonstration I observed that the three-legged stool with the jerrycan placed at its far corner toppled when nudged, while the wider four-legged stool with the same load at its corner only rocked and returned to its upright position. I also observed from the animation that the vertical line through the centre of gravity crossed outside the	I learned that an object topples when the vertical line through its combined centre of gravity falls outside the base of support, creating an unbalanced moment that rotates the object away from equilibrium. I also learned that stability is increased by lowering the centre of gravity and by widening the base of support, and that	The Jiko stool topples because placing the heavy jerrycan at its far corner shifts the combined centre of gravity of the stool-plus-load system outward. When the vertical line through that combined centre of gravity moves beyond the edge of the base of support — the triangle formed by the three legs — the clockwise moment of the load

	base of support at the exact moment the narrower stool committed to toppling.	engineers deliberately use both strategies in structures such as cranes, double-decker buses, and grain silos. I can now calculate moments and use the Principle of Moments to predict whether a loaded object will topple or remain balanced.	about the nearest leg is greater than the anticlockwise moment of the rest of the stool. This net unbalanced moment rotates the stool until it falls. The wider stool does not topple because its larger base of support keeps the vertical line through the centre of gravity inside the triangle even when the load is near the edge, so the net moment remains balanced or self-correcting. The same physics explains why engineers lower the centre of gravity and widen the base in every structure that must resist toppling forces.
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ATTRIBUTION AND LICENSING

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